



CUSTOMER ANALYTICAL REPORT

Quang Phuoc Dao

quangphuoc.work@gmail.com

CONTENTS

I. OVERVIEW	3
1. Introducing to Customer 360 Analytics	3
2. RFM Model	4
3. IQR.....	5
4. BCG Matrix	5
II. CUSTOMER SEGMENTATION PROCESS	7
1. Data information	7
2. RFM Calculation	7
3. Score RFM.....	8
4. Segment customers.....	9
III. INSIGHTS & RECOMMENDATIONS:	10
1. Overview	10
2. Finding Insights	10
3. Sharing Recommendations.....	12
IV. SQL CODE REFERENCE:	14

I. OVERVIEW

1. Introducing to Customer 360 Analytics

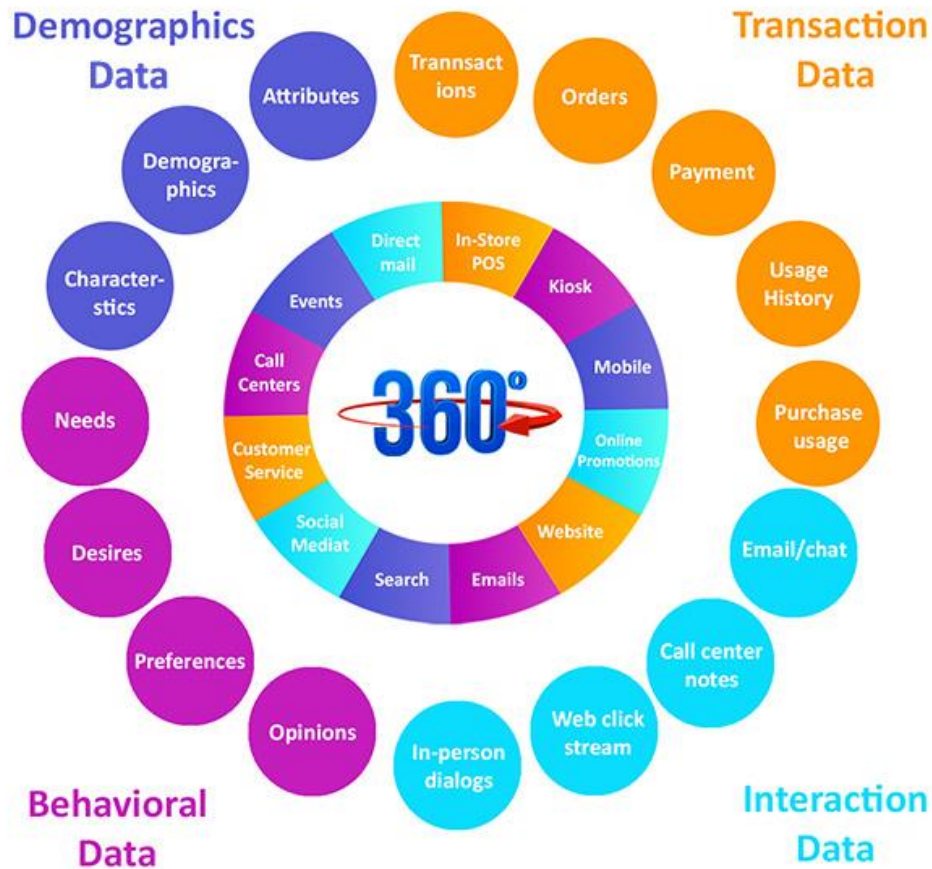


Image 1: Customer 360 Analytics

Customer 360 Analytics is a unified approach to understanding customers by consolidating data from multiple touchpoints and channels. By creating a single, integrated view of each customer, businesses can derive actionable insights that drive improved customer experience, engagement, and operational efficiency. Here's a detailed breakdown of its components, benefits, and applications:

Demographic Data: Mainly about basic customer information such as: Age, gender, and location...

Transactional Data: Insights derived from customers' purchase history and financial interactions: Purchase frequency, order values, and lifetime value...

Interaction Data: Data from customer touchpoints across multiple channels:

Engagements with customer service (e.g., email, chat, calls), social media mentions and reviews

Behavioral Data: Contextual data capturing customer preferences, opinions, and intent: Browsing patterns on websites and apps, responses to surveys, polls, or other feedback tools

2. RFM Model

RFM analysis is a technique used to quantitatively rank and group customers based on the recency, frequency and monetary total of their recent transactions to identify the best customers and perform targeted marketing campaigns.

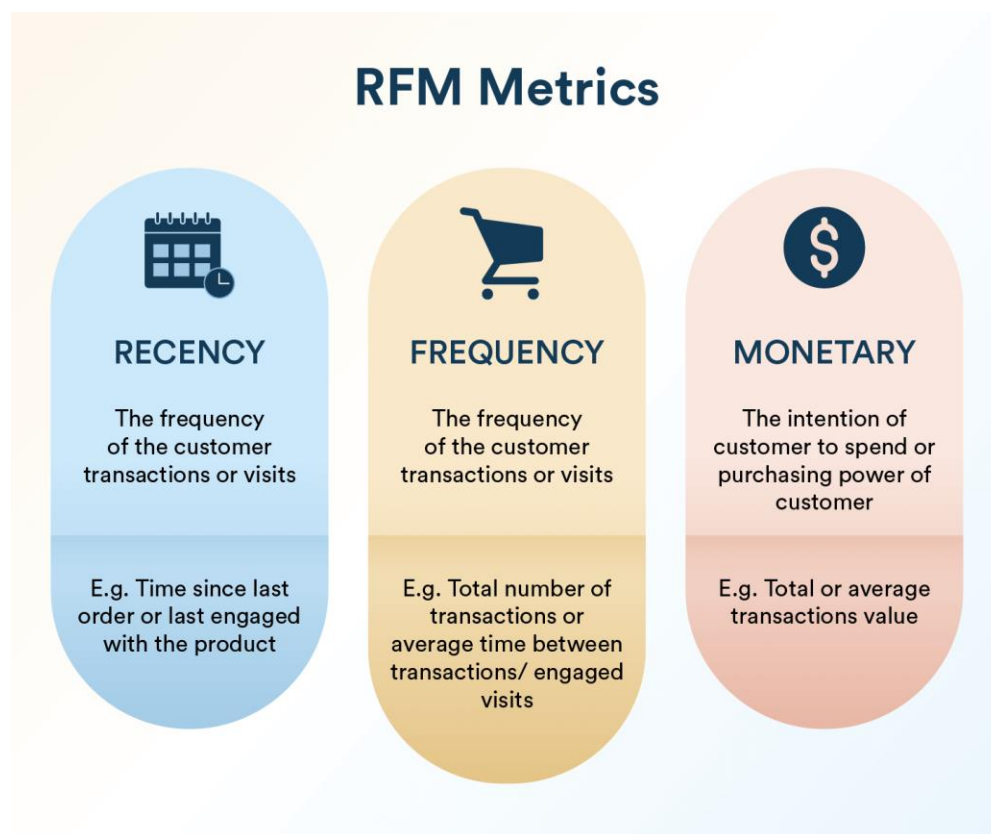


Image 2: RFM Metrics

Recency: the time between customers' last use of service and the report date.

Frequency: the average rate at which customers use the service in the reported period.

Monetary: total amount of revenue receive from each customer in the reported period.

3. IQR

Interquartile Range (IQR) is a statistical measure that is used to measure the variability of a dataset. It is a measure of the spread of the middle 50% of the data and is less affected by extreme values or outliers

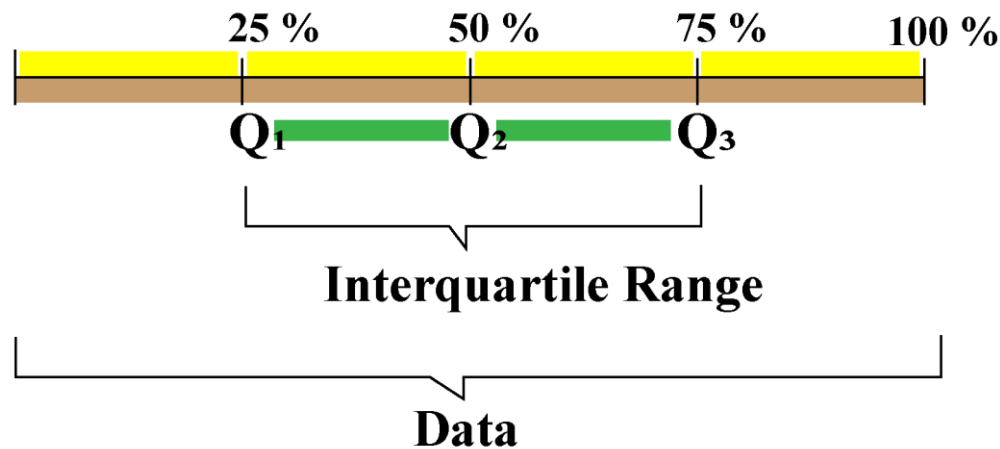


Image 3: IQR

IQR method is used to calculate RFM points. The overall process is:

- Sorting data in ascending/descending order depends on the meaning of the data
- Getting the MEDIAN (Q₂), Q₁ & Q₃ base on the position (row number)
- Assign data into 4 levels: 1 is less/more than Q₁, 2 is between Q₁ & Q₂, 3 is between Q₂ & Q₃, and 4 is the rest.

This will be applied to all 3 measures from the RFM model. For Recency, the smaller the better, so it should be sorted in descending order while the other 2 are in ascending order

4. BCG Matrix

The Boston Consulting Group (BCG) growth-share matrix is a planning tool that uses graphical representations of a company's products and services in an effort to help the company decide what it should keep, sell, or invest more in.

BCG MATRIX

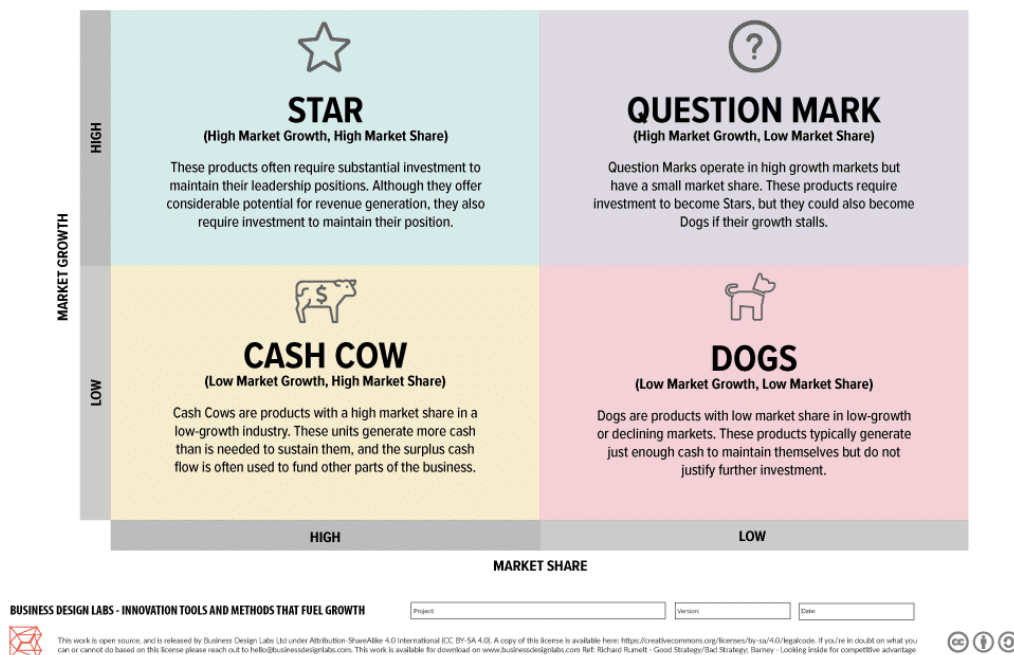


Image 4: BCG Matrix

The BCG growth-share matrix breaks down products into four categories, known simply as dogs, cash cows, stars, and question mark

- **Low Growth, High Share:** Companies should milk these “cash cows” for cash to reinvest.
- **High Growth, High Share:** Companies should significantly invest in these “stars” as they have high future potential.
- **High Growth, Low Share:** Companies should invest in or discard these “question marks,” depending on their chances of becoming stars.
- **Low Share, Low Growth:** Companies should liquidate, divest, or reposition these “pets.”

→ **This matrix** will be applied to classify all customers into **4 major groups** based on **RFM point**: Vip Customers, Loyal Customers, New Customers, Hibernated/Inactive Customers

II. CUSTOMER SEGMENTATION PROCESS

1. Data information

In this case, we have two tables that need to be analyzed: Customer_Transaction table, Customer_Registered table: contain more than 100,000 customers within 3 months from 1/6/2022 to 1/9/2022.

Column_name	Data type
ID	Bigint
Contract	Varchar
LocationID	Int
BranchCode	Int
Status	Int
Create_Date	Datetime
Stop_Date	Datetime

Table 1. Customer_Registered

Column_name	Data type
ID	Bigint
CustomerID	Varchar
Purchase_Date	Datetime
GMV	Bigint

Table 2. Customer_Tansaction

2. RFM Calculation

Recency	The customer's last purchase (use September 1, 2022 to calculate)
Frequency	How often a customer makes a purchase within a specific period
Monetary	Total amount a customer has purchased/Lifetime

ABC customerid	123 recency	123 frequency	123 monetary	123 rn_recency	123 rn_frequency	123 rn_monetary
158921	31	0.14	40,584	12,290	113,179	4,951
91984	31	0.14	40,584	11,802	113,346	4,952
93368	31	0.14	40,584	11,856	113,647	4,953
181164	31	0.14	40,584	12,801	114,073	4,954
289879	92	0.17	40,378	74,688	106,557	4,958
283281	92	0.17	40,378	74,285	106,154	4,959
139204	92	0.14	40,000	78,718	110,618	5,072
92430	62	0.14	39,870	54,986	111,862	5,073
213965	92	0.17	39,166	77,003	108,872	5,557

Table 3. Calculating Recency, Frequency, Monetary

3. Score RFM

ABC customerid	123 recency	123 frequency	123 monetary	ABC r	ABC f	ABC m
765319	6	0.25	0	4	2	1
651466	4	0.25	0	4	2	1
847826	3	0.25	0	4	2	1
1008506	4	0.25	0	4	2	1
916721	4	0.25	0	4	2	1
321377	1	0.17	376	4	1	1
245929	1	0.17	403	4	1	1
286639	1	0.17	430	4	1	1
227237	31	0.17	457	3	1	1
546235	31	0.2	483	3	1	1

Table 4: RFM Calculating according to IQR

IQR is calculated as below table:

IQR	Min	Q1	Q2	Q3	Max
Recency (Days)	1	31	62	92	92
Frequency (times/years)	0.14	0.19	0.23	0.26	1.2
Monetary(VND/years)	0	15.625	19,712.69	23,831.62	152,939.85

Table 5: RFM range according to IQR

Then, RFM scoring is measured in the scale from 1 to 4:

Score	1	2	3	4
R	≥ 92	92-62	62-31	31-1
F	≤ 0.2	0.19-0.23	0.23-0.26	≥ 0.26
M	$\leq 15,625$	15,625-19,712.69	19,712.69 – 23,831.62	$\geq 23,831.62$

4. Segment customers

Type	Group	Characteristics	RFM
Star	VIP Customers	Use the service often up to recently, spend a decent amount of money	444,443,434
Cash Cow	Loyal Customers	Use the service regularly, not necessary recent, spend decent amount of money	441,431,442,342,133 243,143,233,134,234 144,244
Question Mark	New Customers	Recently use the service, may not many times but not too few, pay reasonable amount of money	411,311,421,412,422 413,423,414,424
Dogs	Hibernated Customers	Don't use the service often and recently, pay little money	141,241,131,211,111 221,121,222,212,242 112,142,114,132,232 223,123,231,213,113 224,12

III. INSIGHTS & RECOMMENDATIONS:

1. Overview

114.08K

Count of CustomerID

61.25

Average of recency

0.25

Average of frequency

21.58K

Average of monetary

Image 5: Business overview

From the information gathered, there was 114.08 thousand transactions made by registered customers from the 1st of June until the 1st of September 2022.

On average, a customer will make a transaction within 60 days, the average number of purchases was 0.25 times a year with the value being 21580 VND

2. Finding Insights

Customer Segmentation based on RFM Result

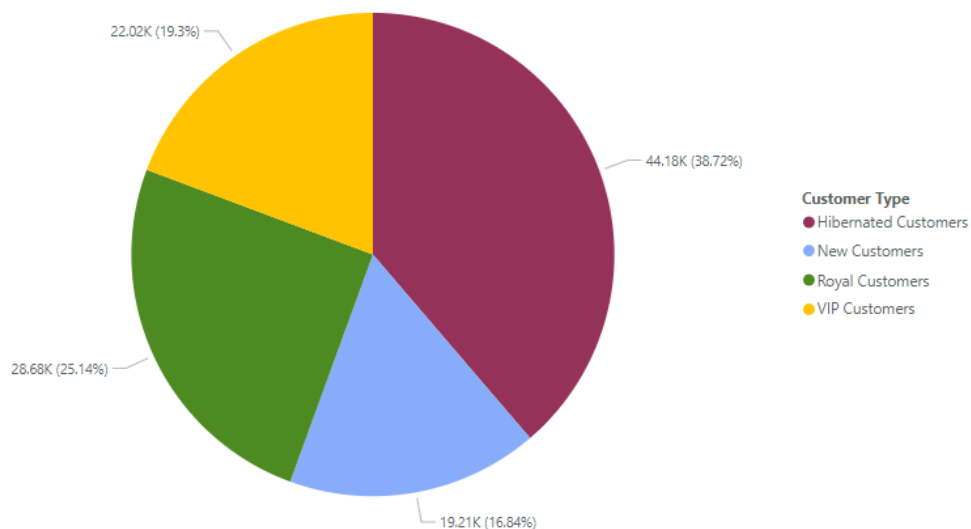


Image 6: Proportion of each group

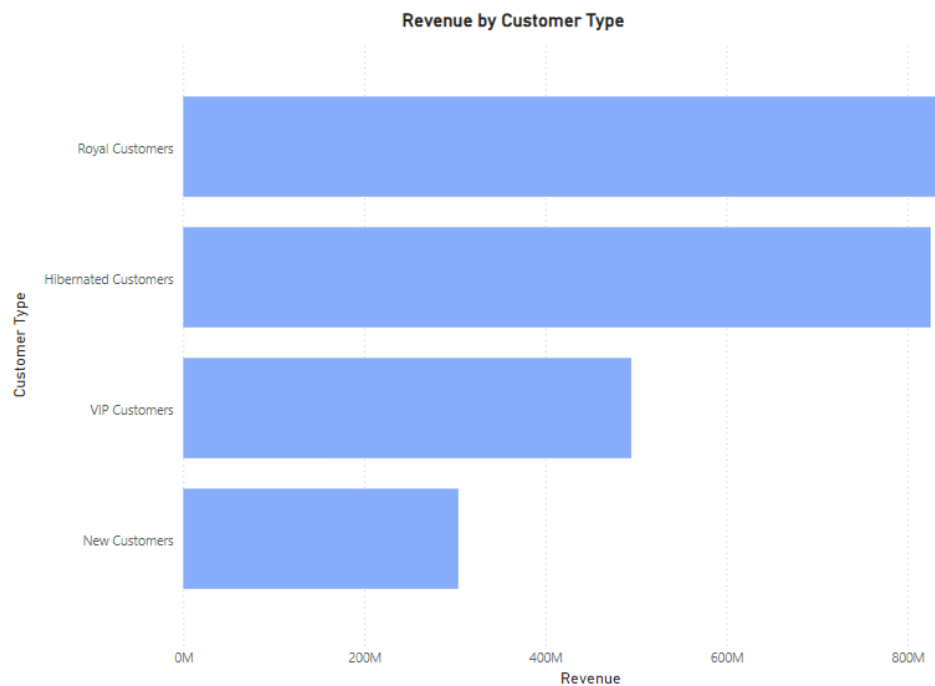


Image 7: Revenue by each group

- **VIP Customers (19.3%):** Although VIP Customers comprise only 19.3% of the customer base, they generate an impressive 500 million in revenue. Their spending behavior is characterized by high frequency and significant amounts, making them the most profitable group on a per-customer basis. This segment is critical for the business's profitability and long-term success. The relatively smaller size of this group compared to others suggests that they represent the elite tier of customers who are highly engaged with the service. Retaining this group and continuing to offer exclusive experiences and benefits will be essential to maintaining their strong revenue contribution.

- **Royal Customers (25.14%):** Royal Customers represent 25.14% of the customer base and are the largest contributors to revenue, generating around 800 million. These customers use the service regularly, though their activity may not always be recent. Their consistent engagement and reliable spending patterns indicate a stable and loyal foundation for the business. Despite being smaller in size compared to the Hibernated Customers, their higher per-customer spending highlights their importance. This group's high revenue contribution also points to their potential as advocates for the brand if nurtured effectively through personalized rewards and engagement programs.

- **New Customers (16.84%):** New Customers are the smallest segment, accounting for 16.84% of the customer base and contributing approximately 300 million in revenue. As these customers are in the early stages of their relationship with the service, their spending levels and frequency of engagement are still developing. This group reflects the business's ability to attract new users but also highlights the need for effective onboarding strategies to encourage them to transition into higher-value segments like Royal or VIP Customers. Their lower revenue contribution is expected at this stage, but the segment is critical for ensuring a healthy pipeline of future loyal customers.

- **Hibernated Customers (38.72%):** This is the largest customer segment, making up nearly 40% of the total customer base. These customers are characterized by infrequent usage of the service, low spending, and a lack of recent engagement. Despite their substantial size, their revenue contribution stands at approximately 700 million, reflecting low average spending per customer. The combination of high volume and moderate revenue underscores a significant untapped potential. A large portion of the customer base is disengaged, which may indicate gaps in customer retention strategies or areas where customer satisfaction needs improvement. This segment presents an opportunity for reactivation campaigns aimed at converting them into more active and profitable groups.

3. Sharing Recommendations

- **VIP Customers (19.3% | ~500M Revenue):**
 - Create VIP-only benefits, such as early access to new products, private events, or personalized service experiences.
 - Assign dedicated account managers or personalized concierge services to ensure their needs are met promptly and effectively.
 - Encourage these highly engaged customers to become brand ambassadors by offering referral incentives or spotlighting them in promotional campaigns.
- **Royal Customers (25.14% | ~800M Revenue):**
 - Introduce a robust rewards program offering exclusive benefits, such as discounts, free upgrades, or access to premium services. This can solidify their loyalty and keep them engaged.

- Send personalized messages through newsletters or notifications about new products, services, or events tailored to their preferences.
- Leverage their established trust in the brand by offering complementary services or products that align with their past usage patterns.

- **New Customers (16.84% | ~300M Revenue):**

- Develop a structured onboarding process to educate new customers about the service, including tutorials, welcome offers, or walkthroughs.
- Send personalized messages through newsletters or notifications about new products, services, or events tailored to their preferences.
- Leverage their established trust in the brand by offering complementary services or products that align with their past usage patterns.

- **Royal Customers (25.14% | ~800M Revenue):**

- Design targeted campaigns, such as personalized emails or SMS, offering discounts, free trials, or special promotions to encourage these customers to re-engage with the service.
- Implement surveys or direct outreach to understand the reasons for disengagement and address their pain points. This could involve improving service quality, adjusting pricing, or introducing new features.
- Incorporate loyalty programs that reward customers for renewed activity, such as points for purchases or discounts after a set number of transactions.

IV. SQL CODE REFERENCE

```

with basic_rfm as (
  select
    customerid,
    datediff(day, max(cast(purchase_date as date)), '2022-09-01') as recency,
    round(
      cast(count(distinct cast(purchase_date as date)) as float) /
      cast(nullif(datediff(year, cast(created_date as date), '2022-09-01'), 0) as float),
      2
    ) as frequency,
    sum(gmv) / nullif(datediff(year, cast(created_date as date), '2022-09-01'), 1) as
monetary
  from
    customer_transaction ct
  join
    customer_registered cr on cr.id = ct.customerid
  where
    customerid != 0
  group by
    ct.customerid, cr.created_date
),

```

```

rn as (
  select
    customerid,
    recency,
    frequency,
    monetary,
    percent_rank() over (order by recency) as rn_recency,
    percent_rank() over (order by frequency) as rn_frequency,
    percent_rank() over (order by monetary) as rn_monetary,

```

```

percentile_cont(0.25) within group (order by recency) over () as r_25,
percentile_cont(0.50) within group (order by recency) over () as r_50,
percentile_cont(0.75) within group (order by recency) over () as r_75,
percentile_cont(0.25) within group (order by frequency) over () as f_25,
percentile_cont(0.50) within group (order by frequency) over () as f_50,
percentile_cont(0.75) within group (order by frequency) over () as f_75,
percentile_cont(0.25) within group (order by monetary) over () as m_25,
percentile_cont(0.50) within group (order by monetary) over () as m_50,
percentile_cont(0.75) within group (order by monetary) over () as m_75
from
    basic_rfm
),

rfm as (
    select
        customerid,
        recency,
        frequency,
        monetary,
        case
            when recency >= r_75 then '1'
            when recency >= r_50 and recency < r_75 then '2'
            when recency >= r_25 and recency < r_50 then '3'
            else '4'
        end as r,
        case
            when frequency <= f_25 then '1'
            when frequency > f_25 and frequency <= f_50 then '2'
            when frequency > f_50 and frequency <= f_75 then '3'
            else '4'
        end as f,
        case

```

```

    when monetary <= m_25 then '1'
    when monetary > m_25 and monetary <= m_50 then '2'
    when monetary > m_50 and monetary <= m_75 then '3'
    else '4'
end as m
from
    rn
),

segmentation as (
    select
        *,
        case
            when R || F || M in ('444', '443', '434') then 'VIP Customers'
            when R || F || M in (
                '441', '431', '442', '342', '133',
                '243', '143', '233', '134', '234',
                '144', '244'
            ) then 'Royal Customers'
            when R || F || M in (
                '411', '421', '412', '422', '311',
                '413', '423', '414', '424', '321', '312'
            ) then 'New Customers'
            when R || F || M in (
                '141', '241', '131', '211', '111', '121',
                '221', '122', '222', '212', '242', '112',
                '142', '114', '132', '232', '223', '123',
                '231', '213', '113', '224', '124', '214'
            ) then 'Hibernated/Inactive Customers'
        end as customer_segmentation
    from rfm
)

```